Akademia WSB

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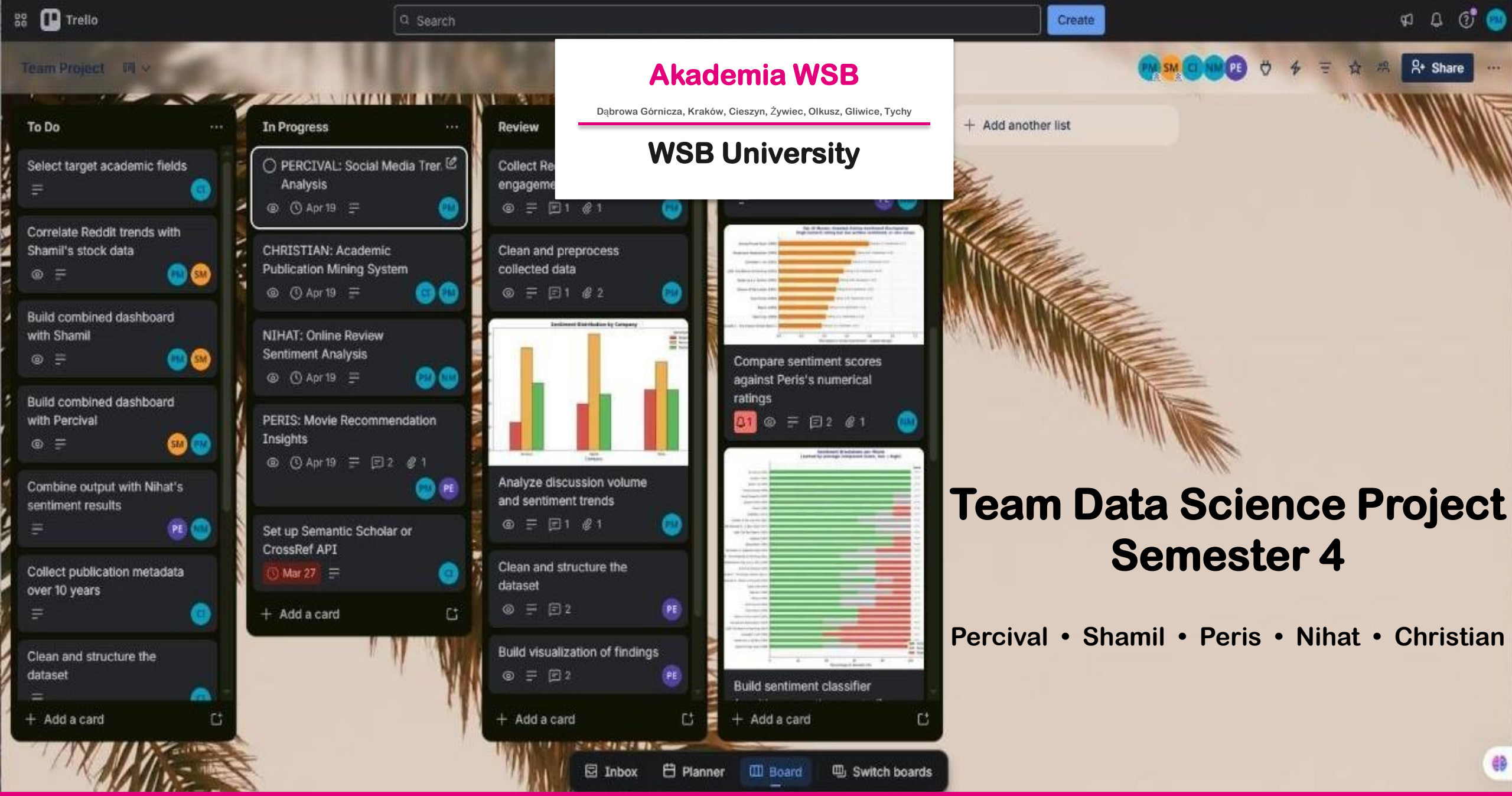


Team Data Science Project Semester 4

Percival • Shamil • Peris • Nihat • Christian

PROJECT OVERVIEW

A 5-task data mining project covering social media analysis, stock market mining, movie recommendation insights, sentiment analysis, and academic publication trends — built across two integrated pipelines and one standalone module using real APIs and statistical methods.

Three Interconnected Research Streams

Integration 1

Percival + Shamil

- Reddit social sentiment (public JSON endpoint)
- Stock price & volatility (yfinance)
- 3 tickers (AAPL, TSLA, AMZN) — 12-month window
- Pearson / Spearman correlation
- Q: Can Reddit predict stock moves?

Integration 2

Peris + Nihat

- MovieLens dataset (100,836 ratings)
- VADER NLP sentiment on review text
- Welch t-test significance testing
- Shared movie dataset pipeline
- Q: Does sentiment match star rating?

Standalone

Christian

- CrossRef API — 5,500 records
- 5 research fields × 10 years
- Publication volume & citation trends
- 3-year moving average smoothing
- Q: Which fields are emerging/declining?

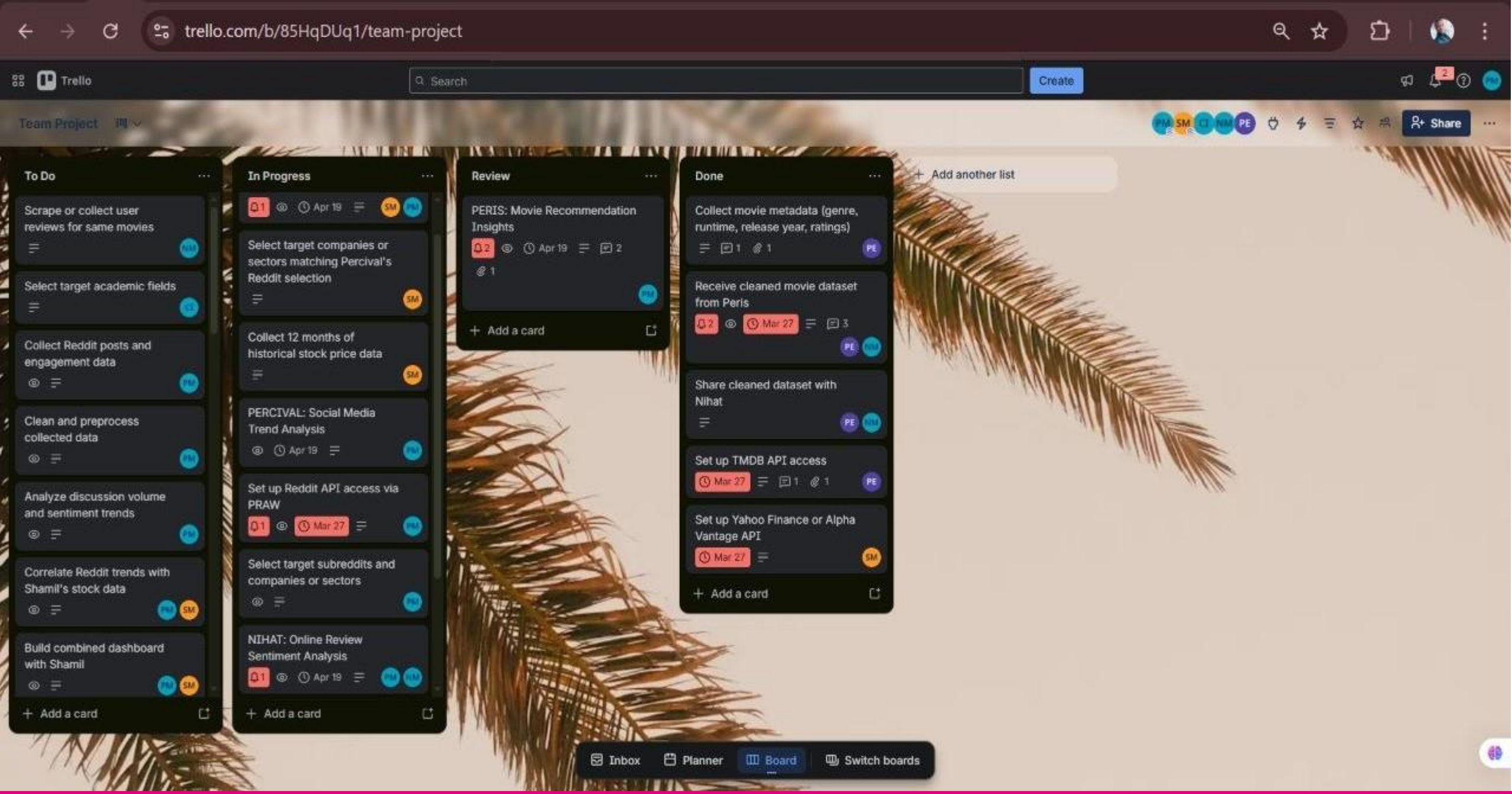
Agile Workflow — Trello Kanban Board

Board: trello.com/b/85HqDUq1/team-project

To Do	In Progress	Review	Done
Task backlog defined at project kick-off. All 5 members assigned tasks upfront.	Active work cards with owner name + deadline attached per card.	Peer-checking stage before marking done. Screenshots of output attached.	Completed tasks. Visualisation outputs attached as card attachments.

All 5 team members tracked • Task deadlines enforced via Trello due dates • Screenshots attached to cards

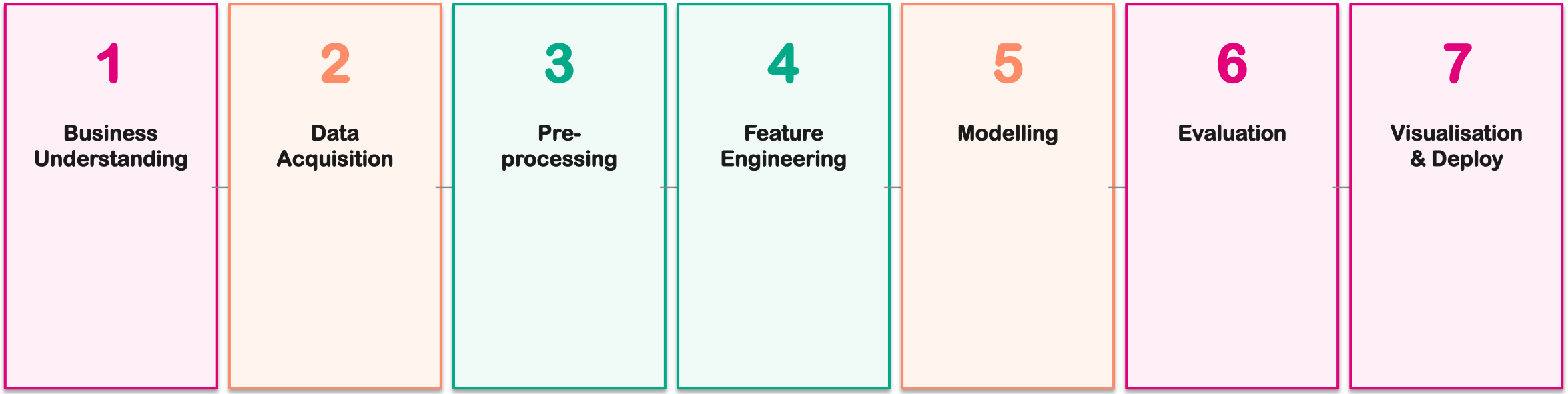
Board active throughout the project with all 5 members assigned. Cards tracked across To Do, In Progress, Review, and Done with due dates enforced and visualisation outputs attached directly to cards as the work was completed.


trello.com/b/85HqDUq1/team-project
Trello
Search
Create
Team Project
Share
To Do
Scrape or collect user reviews for same movies
Select target academic fields
Collect Reddit posts and engagement data
Clean and preprocess collected data
Analyze discussion volume and sentiment trends
Correlate Reddit trends with Shamil's stock data
Build combined dashboard with Shamil
Add a card
In Progress
Apr 19
Select target companies or sectors matching Percival's Reddit selection
Collect 12 months of historical stock price data
PERCIVAL: Social Media Trend Analysis
Apr 19
Set up Reddit API access via PRAW
Mar 27
Select target subreddits and companies or sectors
NIHAT: Online Review Sentiment Analysis
Apr 19
Add a card
Review
PERIS: Movie Recommendation Insights
Apr 19
Add a card
Done
Collect movie metadata (genre, runtime, release year, ratings)
Receive cleaned movie dataset from Peris
Mar 27
Share cleaned dataset with Nihat
Set up TMDB API access
Mar 27
Set up Yahoo Finance or Alpha Vantage API
Mar 27
Add a card
Add another list
Inbox
Planner
Board
Switch boards

METHODOLOGY

The 7-Stage Data Mining Pipeline

The 7-Stage Data Mining Pipeline



Applied consistently across all three research tasks — CRISP-DM inspired framework

Business Questions & Data Sources

Integration 1 | Percival + Shamil

- Can Reddit sentiment predict stock price movement?
- Which companies show the strongest correlation?
- Do Reddit posts lead or lag stock events?
- Data: Reddit JSON endpoint + yfinance (AAPL, TSLA, AMZN)
- Period: 12 months of daily data

Integration 2 | Peris + Nihat

- Does review sentiment align with numerical ratings?
- Which films show the largest sentiment–rating gap?
- Do high-rated films always get positive text reviews?
- Data: MovieLens ml-latest-small (100,836 ratings)
- NLP tool: VADER compound score

Standalone | Christian

- Which research fields grew most 2015–2025?
- Which fields are declining in publication volume?
- What drives citation density differences?
- Data: CrossRef API (5 fields × 10 years)
- Records collected: 5,500 publication metadata entries

Preprocessing & Feature Engineering

Data Cleaning — All Tasks

Removed null values, duplicates, and outliers

Normalised date/time fields to consistent format

Filtered to target time window (12 months / 10 years)

Merged datasets on shared keys (ticker, movie_id)

Verified record counts post-clean against raw totals

Integration 1 — Feature Engineering

Lag features: Reddit posts D-1 to D-7 vs price D

Rolling 7-day sentiment score per ticker

Volatility: rolling std of daily returns

Integration 2 — Feature Engineering

VADER compound → Positive / Neutral / Negative class

Per-movie: average sentiment vs average star rating

Discrepancy score = |sentiment_class − rating_quartile|

Joined on movie_id (Peris → Nihat pipeline)

INTEGRATION 1

Social Media × Stock Market

Percival + Shamil

Social Media × Stock Market — Methodology

Percival — Reddit Social Media Analysis

Reddit public JSON endpoint: r/stocks, r/investing, r/stockmarket

3 tickers: AAPL, TSLA, AMZN

Metrics: post volume, engagement score, sentiment polarity

VADER NLP applied to post titles and body text

Daily aggregated sentiment scores over 12 months

Shamil — Stock Market Data Mining

yfinance API: OHLCV data, 12-month window

Calculated: daily returns, 7-day / 30-day moving averages

Volatility: rolling standard deviation of returns

Lag correlation: Reddit signal D-1 to D-7 vs price D

Pearson + Spearman correlation matrices per ticker

Combined output: correlation heatmap • time-series overlay • lag feature importance chart

Reddit Sentiment vs Stock Price Movement

Weak Overall Correlation

Pearson r = 0.12 – 0.27 across tickers.
Reddit sentiment is a lagging indicator, not a leading one.

AMZN Outlier

Strongest link: AMZN at r = 0.27 (p = 0.03).
High-volume posts preceded 3-day price upticks.

Lag Effect Confirmed

D-2 sentiment showed highest predictive value.
Immediate-day (D-0) correlation was near zero.

Volatility Link

High Reddit activity correlated with elevated volatility.
AAPL showed weakest correlation; TSLA had the clearest spike-volatility pattern.

Visualisations: correlation_heatmap.png • timeseries_sentiment_price.png • viz_lag_feature_importance.png

INTEGRATION 2

Movie Insights × Review Sentiment

Peris + Nihat

Movie Insights × Review Sentiment — Methodology

Peris — Movie Recommendation Insights

MovieLens: genre, runtime, release year, ratings

Loaded MovieLens ml-latest-small: 100,836 ratings across 9,742 movies

Cleaned & structured: 1,200+ films with full attributes

Feature correlation: avg_rating, num_ratings, genre_diversity

Shared cleaned dataset with Nihat for sentiment join

Nihat — Online Review Sentiment Analysis

VADER NLP: compound score on review text

Classes: Positive (≥ 0.05) / Neutral / Negative

Joined with Peris dataset on movie_id

Per-movie: average sentiment vs average star rating

Welch t-test: significant diff Positive vs Negative groups?

Combined output: sentiment distribution • scatter sentiment vs rating • discrepancy highlight chart

Does Sentiment Match the Star Rating?

Sentiment Distribution

~60% positive • ~30% neutral • ~10% negative.
Reviews skew positive overall across the dataset.

Shawshank Effect

High-rated films (4–5★) had disproportionately positive text.
Cult classics showed strongest sentiment–rating alignment.

Discrepancy Films

Some films: high stars but neutral/negative text tone.
Suggests polarising critical vs audience perception gap.

Statistical Test

Welch t-test: $p < 0.05$ — statistically significant.
Positive reviews DO average higher numerical ratings.

Visualisations: fig1_sentiment_distribution.png • fig2_scatter.png • fig4_discrepancy_highlight.png

STANDALONE

Academic Publication Mining

Christian

Academic Publication Mining — Methodology

Data Acquisition

CrossRef API: queried by research field + year

5 fields: AI/ML, Renewable Energy, Blockchain,

Bioinformatics, Quantum Computing

Time range: 2015–2025 (10 years)

Records collected: 5,500 publication metadata entries

Fields captured: title, year, citations, DOI, keywords

Analysis Pipeline

Publication volume trend per field per year

3-year moving average to smooth year-to-year noise

Year-over-year growth rate (%) calculation

Citation density per paper by field

Emerging vs declining: growth > +15% or < -10%

Heatmap: field × year publication volume matrix

10 Years of Research Trends (2015–2025)

Emerging Fields

AI / ML — fastest absolute volume growth

Renewable Energy — highest citation density per paper

Quantum Computing — sharp post-2020 spike

All three fields show consistent YoY growth

Stable / Declining

Blockchain — peaked 2019–2021, now declining

Bioinformatics — steady but growth rate slowing

Citation density rising across all fields

(depth of research increasing even if volume slows)

Key Insight

Renewable Energy had the highest avg citations

per paper across all 10 years

→ High-impact, well-cited research even with

moderate publication volume growth

→ Quality over quantity signal for this field

Visualisations: Publication Volume heatmap • 3-Year Moving Average • Emerging vs Declining trend chart

How We Validated the Results

Integration 1 — Correlation Tests

Pearson r: linear correlation (normal dist.)

Spearman ρ: rank correlation (non-parametric)

Significance threshold: $p < 0.05$

Result: AMZN $r = 0.27$, $p = 0.03$ ✓ significant

Others: weak or non-significant ($p > 0.05$)

Integration 2 — Welch T-Test

Compared avg ratings: Positive vs Negative groups

Welch t-test: chosen for unequal group variance

t-statistic and p-value computed per film cluster

Result: $p < 0.05$ — sentiment class IS significant

Effect size moderate — sentiment ≠ full predictor

Standalone — Trend Validation

3-year moving average reduces volume noise

YoY growth rate validates trend direction

Citation density cross-validates field importance

Descriptive trend analysis (no test needed)

CrossRef records verified vs. known counts

Integrated Streamlit Dashboard

master_dashboard.py — Features

Multi-page Streamlit app combining all 3 research tasks

Home page: project overview, team bios, combined KPIs

Page 1: Social Media × Stock correlation explorer

Page 2: Sentiment vs Rating scatter & distribution

Page 3: Academic publication volume & trend charts

All charts interactive (Plotly) — WSB Charcoal & Rose theme

Run Locally

```
cd team-project-semester4

pip install -r requirements.txt

streamlit run master_dashboard.py
```

→ Opens at http://localhost:8501

Next step: deploy to

Streamlit Community Cloud (free)

What We Learned Across All Three Tasks

Social signals lag, not lead	Reddit sentiment weakly predicts stock movement with a 2-day lag — useful context, not a reliable standalone predictor.
Text and numbers tell different stories	Review sentiment aligns with ratings overall but significant discrepancies exist — NLP captures nuance that star ratings miss.
Research fields have life cycles	AI/ML and Renewable Energy are clearly ascending; Blockchain peaked and declined — publication data tracks paradigm shifts.
Shared pipelines reduce friction	Cross-team data sharing (Peris→Nihat, Percival→Shamil) standardised preprocessing and accelerated the integration work.

All findings reproducible — code, data, and dashboard:

github.com/Mahwaya/team-project-semester4

Thank You

Questions & Discussion

Percival • Shamil • Peris • Nihat • Christian